Solving systems of equations with the [redacted]stitution method

1 Solve the following systems of equations using the substitution method:

a $y = -5x - 23$
$y = 7x + 25$

b $y = 5x + 34$
$y = 3x + 18$

c $y = 5x + 12$
$4y = 28x + 72$

d $y = -4x - 17$
$3y = 21x + 147$

e $y = -6x - 26$
$x + y = -6$

f $y = 3x - 18$
$x - y = 10$

g $y = -5x + 22$
$6x + y = 26$

h $y = 6x - 2$
$-3x - y = -7$

i $y = 8x - 30$
$5x + 9y = 38$

j $y = -2x - 1$
$x + 2y = 13$

k $4x + 3y = 52$
$7x - 5y = 9$

l $y = 3x - 18$
$x + y = -2$

m $y = 5x - 8$
$2x - 3y = -15$

n $2x - 5y = -10$
$y = -5x + 6$

o $y = 2.3x - 17.62$
$y = -8.5x + 30.98$

2 Find the coordinates of the point that the two lines $y = 3x$ and $y = -4x$ have in common.

3 Consider the following system of linear equations:

$$\begin{aligned} -6x - 2y &= -28 \\ 2x + 16y &= 40 \\ 4x - 2y &= 12 \end{aligned}$$

a Find the values of x and y that satisfy the first two equations.

b Determine if this solution satisfies the third equation by substituting the values of x and y into the left hand side of the equation.

c Hence state whether the lines are concurrent.